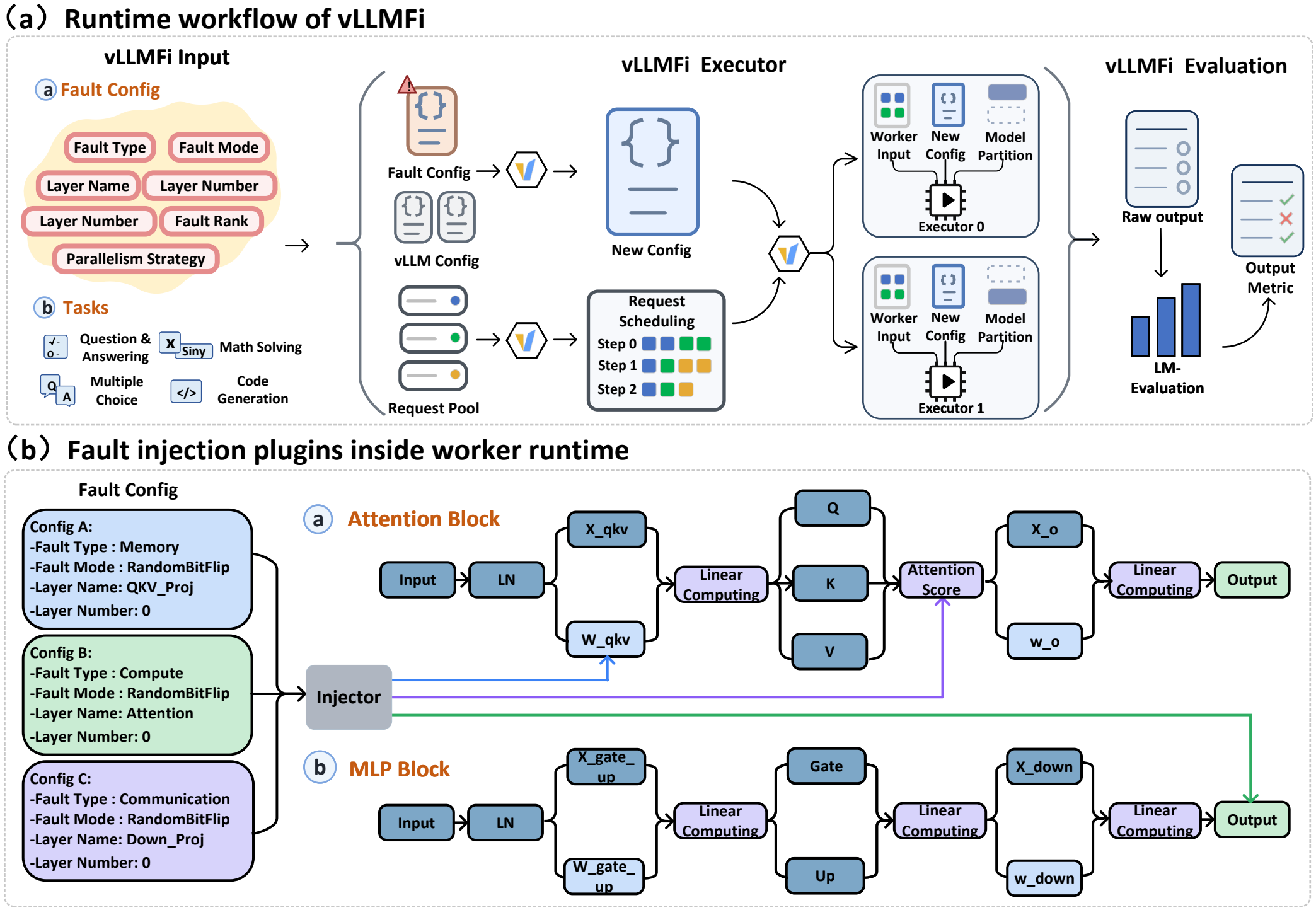

(a) Runtime workflow of vLLMFi
vLLMFi Input
a Fault Config
Fault Type
Fault Mode
Layer Name
Layer Number
Layer Number
Fault Rank
Parallelism Strategy
b Tasks
Question & Answering
Math Solving
Multiple Choice
Code Generation
vLLMFi Executor
Fault Config
vLLM Config
Request Pool
New Config
Request Scheduling
Step 0
Step 1
Step 2
Worker Input
New Config
Model Partition
Executor 0
Executor 1
vLLMFi Evaluation
Raw output
LM-Evaluation
Output Metric
(b) Fault injection plugins inside worker runtime
Fault Config
Config A:
-Fault Type : Memory
-Fault Mode : RandomBitFlip
-Layer Name: QKV_Proj
-Layer Number: 0
Config B:
-Fault Type : Compute
-Fault Mode : RandomBitFlip
-Layer Name: Attention
-Layer Number: 0
Config C:
-Fault Type : Communication
-Fault Mode : RandomBitFlip
-Layer Name: Down_Proj
-Layer Number: 0
Injector
a Attention Block
Input
LN
X_qkv
W_qkv
Linear Computing
Q
K
V
Attention Score
X_o
w_o
Linear Computing
Output
b MLP Block
Input
LN
X_gate_up
W_gate_up
Linear Computing
Gate
Up
Linear Computing
X_down
w_down
Linear Computing
Output